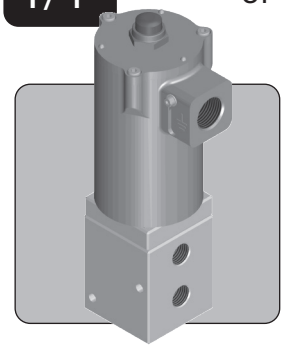
- 2/3DS01 -

DIRECT SOLENOID VALVE

1/4"

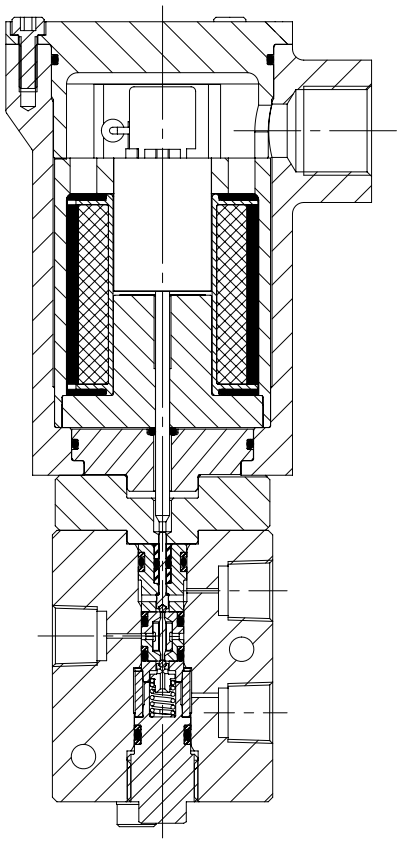
UP TO 690 BAR 10,000 PSI



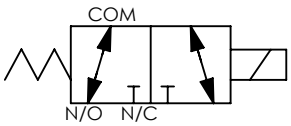
The 2/3DS01 is a 2 position 3 way direct acting stainless steel high pressure solenoid valve. This versatile valve switches high pressures directly via a push style solenoid coil and is ideal for driving small actuators or providing a pressure signal for larger control valves.

A wide variety of low wattage solenoid thrusters are available, including ATEX and IECEx approved coils for use in hazardous areas and numerous electrical connections. Override and reset options provide additional functionality for systems where the ability for manual intervention is required.

- 2 position 3 way direct acting solenoid valve
- Stainless steel construction
- 40 micron filter screen for superior contamination protection
- Efficient 3 watt coil provides high force from low power
- Independently certified SIL 3 capability
- Coil may be rotated 360° to suit cable layout
- Built-in surge suppression diode for fast response
- Wide variety of mounting, connection, voltage, override and reset options available



Specifications

BASIC MODEL NUMBER	2/3AS25
SYMBOL	
MAX WORKING PRESSURE LIQUID	690 bar (10,000 psi)
CV (FLOW CAPACITY)	0.01
FLUID	Liquids only See materials section
TEMPERATURE RANGE	See Product Selector opposite and Technical Data section
PORT SIZE	1/4"
WEIGHT	3.3 kg (7.3 lb)

Specifications may change without notice

Materials

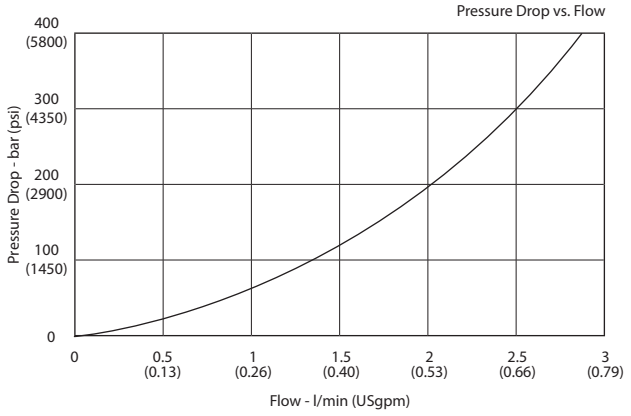
Externally Exposed Parts: 316 Stainless Steel. Acetal push button for manual overide and reset options.

Internally Wetted Parts: 316, 302 and 17-4 PH Stainless Steel, Silicon Nitride, Peek and Acetal.

The standard valve is designed for use with mineral oils, water glycols and plain water may also be used with a wide variety of media compatible with the materials of construction.

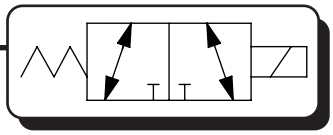
⚠ The standard valve has Viton® seals. Further seal options are available via the Product Selector. Compatibility with the working fluid at the operating temperature must be considered.

Typical Performance



Typical performance based on water

Valve & Thruster Options



Valve Options

As standard the valve will move to the energised position when a suitable electrical supply is applied and is returned by a spring (auto-reset) when the electrical supply is removed.

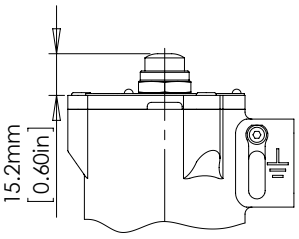
Alternative options include –

MO : Manual Override – With the electrical supply de-energised the valve may be temporarily switched via a push button. Releasing the push button will allow the valve to return. Contact us if a detented manual override feature is required.

MR : Manual Reset – Electrically energising the solenoid will not cause the valve to switch until the push button is pressed. Once the valve is switched the push button may be released and the valve will remain switched. Removing the electrical supply will cause the valve to return to the de-energised position. The valve will not switch if the push button is pressed with the coil de-energised.

MOMR : Manual Override and Manual Reset - Electrically energising the solenoid will not cause the valve to switch until the push button is pressed. Once the valve is switched the push button may be released and the valve will remain switched. Removing the electrical supply will cause the valve to return to the de-energised position. The valve will switch if the push button is pressed with the coil de-energised.

ADDITIONAL HEIGHT FOR MO, MR AND MOMR VARIANTS



Please contact us for further override, reset and latching options.

Thruster & Seal Options

For use in zone 1 and 2 explosive atmospheres as defined by 1999/92/EC either the STEXd or STEXm thruster must be selected. It is the customer's responsibility to assess the application and to determine the zone and temperature class for their particular atmosphere. Please contact us if use in zone 0 is required.

The temperatures stated for seal options relate to the temperature of the fluid inside the valve. The minimum allowable ambient temperature is equal to the minimum allowable seal temperature. The table below details the maximum allowable ambient temperature limits. All coils use class H insulation.

The thruster options included on this data sheet represent only a few common configurations. Please contact us where alternative options are required, such as dual coil windings, alternative electrical supplies, connections and conduits or when subsea use is required.

Thruster Type	Maximum Ambient Temperature for non-hazardous area	Maximum Ambient Temperature for T4 temperature rating	Maximum Ambient Temperature for T6 temperature rating	IP Rating
STHC	90°C	Not allowed	Not allowed	65
STKC	90°C	Not allowed	Not allowed	68
STEXd	90°C	90°C*	65°C	68
STEXm	90°C	80°C*	Not allowed	68

* A maximum ambient temperature of 65°C is allowed if NBR seals are selected

Voltage Option	Voltage Range	Dropout Voltage
12VDC	10.8 – 13.2V	1.2 – 3.0V
24VDC	21.6 – 26.4V	2.4 – 6.0V
48VDC	43.2 – 52.8V	4.8 – 12.0V
115VAC	100.0 – 127.0V 50/60Hz	11.5 – 29.0V
230VAC	215.0 – 253.0V 50/60Hz	23.0 – 57.5V

Product Selector

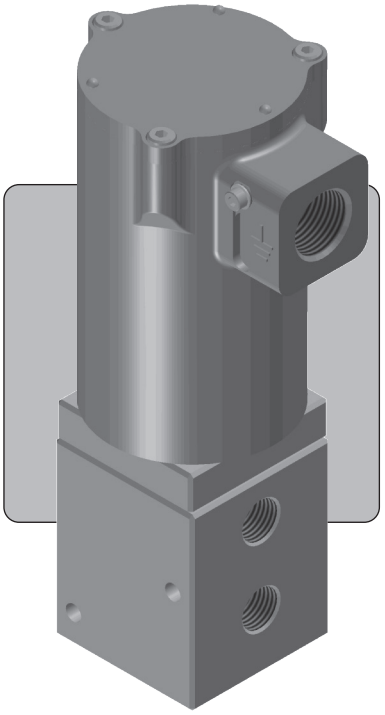
2/3DS01 P - MO - STHC - 12VDC - NBR - 10K

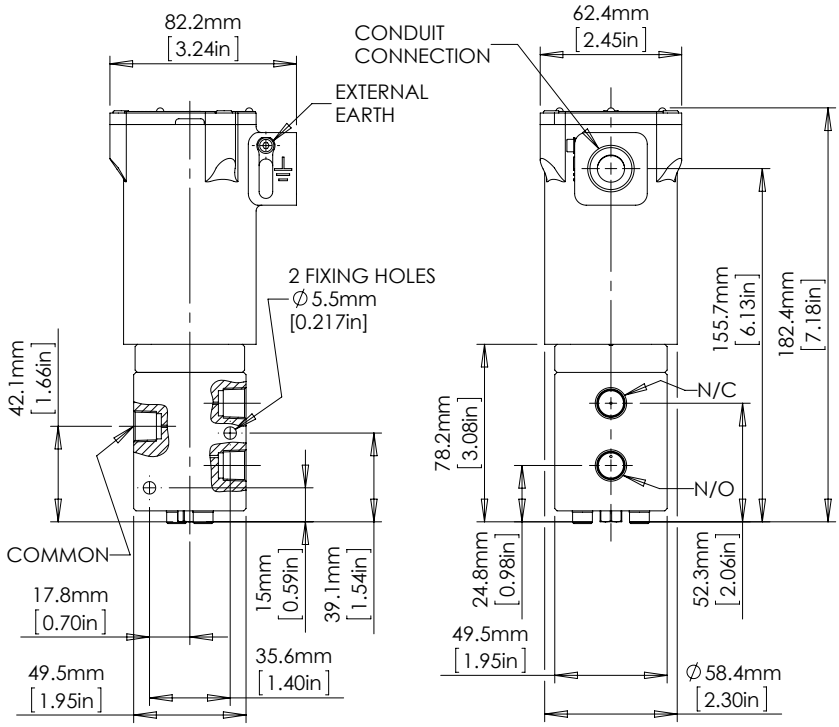
VALVE TYPE	PORTING OPTIONS	VALVE OPTIONS	THRUSTER OPTIONS	VOLTAGE OPTIONS	SEAL OPTIONS	MAX WORKING PRESSURE
2/3DS01	P 1/4" BSPP female N 1/4" NPT female M Manifold mount	Leave blank if none required MO Manual override MR Manual reset MOMR Manual override and manual reset	STHC Din connector STKC M20 conduit STEXd Ex db approved STEXm Ex emb appoved	12VDC 12 volt DC 24VDC 24 volt DC 48VDC 48 volt DC 115VAC 110/120 volt AC 230VAC 230/240 volt AC	Leave blank for Viton® seals -10˚C to +120˚C NBR Nitrile seals -25˚C to +105˚C EP EPDM seals -45˚C to +120˚C LNBR Low temperature Nitrile seals -55˚C to +120˚C HNBR Hydrogenated Nitrile seals -20˚C to +120˚C	10K 690 bar (10,000 psi) Liquid
Repair and seal kits are available on request For a seal kit add SK at the end of the model code and for a repair kit add RK	Further porting options are available on request	See valve and thruster options above for further details	Further thruster options are available on request	STEXm thruster is not available with AC coils Further voltage options are available on request	Further seal options are available on request. Temperature ratings relate to fluid temperature. See above for ambient temperatures	

2/3DS01

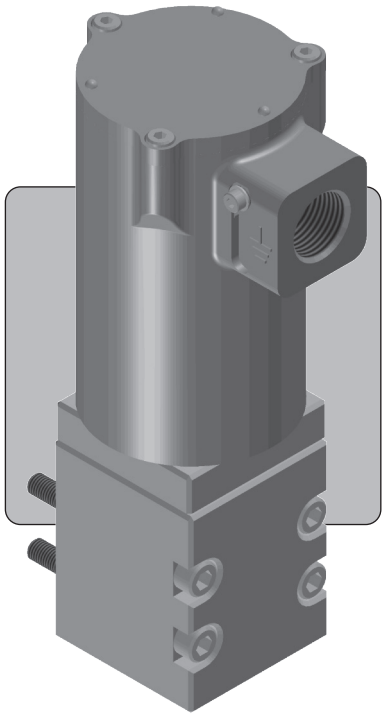
DIRECT SOLENOID VALVE

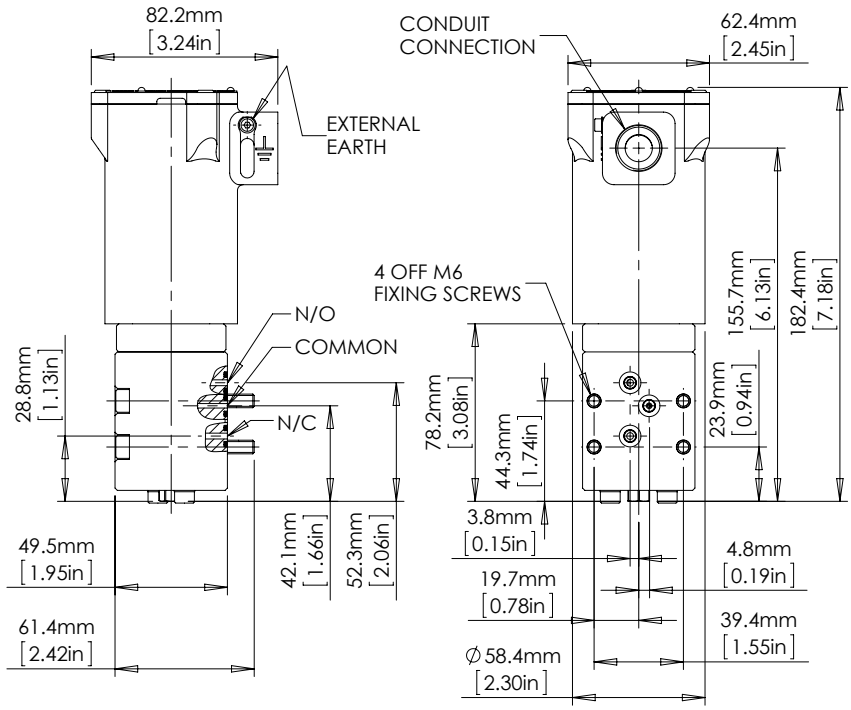
2/3DS01-STEXd



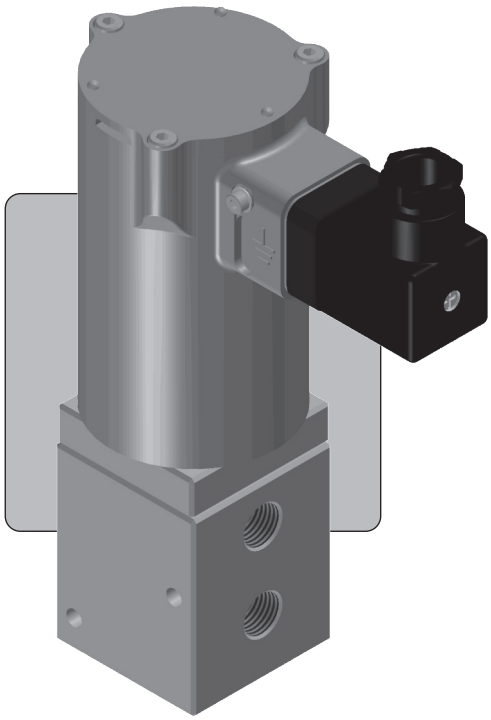


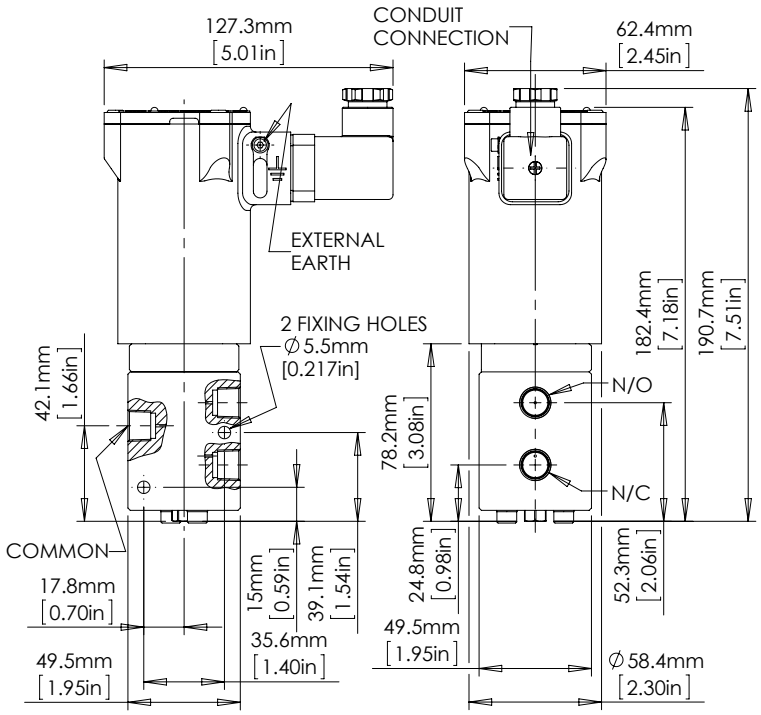
2/3DS01-STEXm





2/3DS01-HC





2/3DS01-KC

